

Revision Quick-Notes: Ruby on Rails (MSE 2)

Unit 3: Rails Models & User Management

1. ACTIVERECORD & MIGRATIONS

- **ActiveRecord (ORM):** Bridge between Ruby and Database.
Map: Table → Class | Row → Object | Column → Attribute.
- **Migrations:** DSL to manage schema versioning.
- **Commands:** `rails db:migrate` (apply), `rails db:rollback` (undo), `rails db:migrate:status` (check).

2. VALIDATIONS & CALLBACKS

- **Validations:** Ensure data correctness before saving.
Helpers: `presence`, `length`, `uniqueness`, `numericality`, `format`.
- **Callbacks:** Automated hooks in object lifecycle.
Hooks: `before_save`, `after_create`, `before_destroy`.

3. MODEL ASSOCIATIONS

- **One-to-Many:** `has_many :posts` / `belongs_to :user`.
- **Many-to-Many:** Uses join table + `has_many :tags, through: :post_tags`.
- **One-to-One:** `has_one :profile` / `belongs_to :user`.
- **Option:** `dependent: :destroy` (deletes children if parent is deleted).

4. USER MANAGEMENT (DEVISE)

- **Purpose:** Authentication (signup, login, sessions).
- **Helpers:** `current_user` (get object), `user_signed_in?` (bool), `authenticate_user!` (force login).

Unit 4: PostgreSQL & SQL Basics

1. POSTGRESQL ARCHITECTURE

- **Client-Server:** Postmaster (Supervisor) manages Backend Processes (one per connection).
- **Memory Structure:** Shared Buffers (Data Cache) | WAL Buffers (Transaction Logs).
- **MVCC:** Multi-Version Concurrency Control (Readers don't block Writers).
- **WAL (Write-Ahead Logging):** Write to logs before data files for durability.

2. ACID PROPERTIES

- **A:** Atomicity (All or nothing).
- **C:** Consistency (Valid state).
- **I:** Isolation (Don't interfere).
- **D:** Durability (Never lost - via WAL).

3. NUMERIC & STRING DATA TYPES

- **Numeric:** `INT`, `DECIMAL`, `SERIAL` (auto-incrementing ID).
- **String:** `VARCHAR(n)` (limited), `TEXT` (preferred, unlimited).
- **Others:** `JSONB` (indexed JSON), `TIMESTAMPTZ` (timezone aware).

4. SQL CRUD & JOINS

- **CRUD:** `INSERT INTO`, `SELECT`, `UPDATE ... SET`, `DELETE FROM`.
CRITICAL: Always use `WHERE` with `UPDATE/DELETE`.
- **Joins:**
 - `INNER JOIN`: Only matches in BOTH.
 - `LEFT JOIN`: ALL Left table rows + Matches from Right.
 - `FULL OUTER JOIN`: ALL rows from Both tables.